Expense Intelligence Dashboard

This content explains the 10 problems addressed by your project, the Power BI visual used for each problem, the relevant dataset insights, and how the dashboard helps solve each problem.

The project follows three stages: **data cleaning and feature engineering in Python/Jupyter Notebook, followed by interactive visualisation in Power BI.** The dashboard consists of three pages: Financial Pulse, Spending Behaviour, and Expense Intelligence.

Problem 1: What is the overall financial position of the household?

Problem statement: It is difficult to understand the household's financial health when income, expenses, and transfers are recorded across many individual transactions.

Solution: The Financial Pulse dashboard summarises total income, total expenses, and net savings through KPI cards.

Chart/visual used:

- Card — Total Income
- Card — Total Expenses
- Card — Net Savings

Data insights:

Financial indicator	Value
Total income	■30,42,397.35
Total expenses	■19,55,380.53
Net savings	■10,87,016.82
Transfer-Out amount	■17,64,780.90

Calculation:

Net Savings = Total Income − Total Expenses

= ■30,42,397.35 − ■19,55,380.53

= ■10,87,016.82

How it solves the problem: The KPI cards give users an immediate summary of the household's financial position without manually adding individual transactions.

Problem 2: How does household spending change over time?

Problem statement: Users may not easily identify months with high expenses or understand whether spending is increasing or decreasing over time.

Solution: The Financial Pulse page uses a monthly spending trend visual to compare expenses across months.

Chart used: Line chart or clustered column chart

- X-axis: `Month_Year_Date`

- Y-axis: Sum of `Expense_Amount`

Data insights: The monthly chart allows the user to identify:

- Months with the highest spending.
- Months with comparatively lower spending.
- Changes in spending across years.
- Periods that may require closer investigation.

How it solves the problem: The time-series chart reveals spending patterns that are difficult to identify from transaction-level records. The date slicer lets the user examine a particular year or period.

Problem 3: Which categories account for the largest expenses?

Problem statement: A large number of transactions makes it difficult to determine which categories consume the most money.

Solution: The Spending Behaviour dashboard compares total expenses across categories.

Chart used: Horizontal bar chart

- Y-axis: `Category`
- X-axis: Sum of `Amount` or `Expense_Amount`
- Sort: Descending by expense amount
- Filter: Transaction type = Expense

For regular-consumption analysis, exclude Money transfer, Investment, Public Provident Fund, and Recurring Deposit.

Data insights:

Normal-consumption category	**Total expense**
Transportation	■1,69,053.78
Household	■1,61,645.58
Subscription	■1,14,587.91
Food	■96,393.10
Other	■87,025.28
Normal-consumption category	**Total expense**
Tourism	■63,608.85
Gift	■40,168.00
Apparel	■25,373.82

Insight: Transportation and Household are the two largest categories among the categories retained for normal-consumption analysis. Subscription and Food also represent substantial expense amounts.

How it solves the problem: The ranked bar chart makes the largest spending areas immediately visible, helping users decide which categories deserve closer review.

Problem 4: Which payment methods are used most frequently?

Problem statement: Users may not know which payment methods they rely on most often or how their payment habits are distributed.

Solution: The Spending Behaviour page compares transaction counts by payment mode.

Chart used: Bar chart or donut chart

- Category/legend: `Mode`
- Values: Count of transactions

Data insights:

Payment mode	Transaction count
Saving Bank account	1,223
Cash	1,046
Credit Card	162

How it solves the problem: The visual reveals payment preferences and makes it easier to compare payment frequency. A separate amount-based chart can be used if the objective is to identify which payment mode accounts for the most spending.

Problem 5: How much money is spent on regular household consumption?

Problem statement: Total expense figures can be misleading when they include transfers, investments, and savings-related transactions alongside ordinary purchases.

Solution: The Spending Behaviour page separates regular-consumption categories from transfers and savings/investment movements.

Chart used: Horizontal bar chart or stacked column chart

- Category: `Category`
- Value: Sum of `Expense_Amount`
- Filter out: Money transfer, Investment, Public Provident Fund, Recurring Deposit

How it solves the problem: Excluding transfer and investment categories produces a more meaningful view of ordinary household spending. It helps users focus on everyday consumption instead of treating every money movement as a purchase.

Problem 6: What proportion of discretionary spending comes from Food and Shopping?

Problem statement: Users may want to understand how much of their flexible spending is associated with Food and Shopping so they can identify opportunities to reduce expenses.

Solution: The Spending Behaviour page compares Food and Shopping with total discretionary spending.

Chart/visual used:

- Donut chart or bar chart — discretionary spending by category.
- KPI card — combined Food and Shopping share.

Calculation:

Food + Shopping Share = ((Food Expense + Shopping Expense) / Total Discretionary Expense) × 100

How it solves the problem: The percentage shows how much of discretionary spending is concentrated in these categories, helping users assess whether reductions in food-related or shopping expenses could improve their budget.

Problem 7: Which spending categories exceed the planned budget?

Problem statement: Without comparing actual expenses against planned limits, users cannot identify which categories need budget adjustments.

Solution: The Spending Behaviour page compares actual spending with a budget amount for each category.

Chart/visual used:

- **Clustered bar chart:** Actual expense versus budget.
- **Table or matrix:** Category, Budget, Actual Expense, Variance, Status.
- **Optional KPI card:** Number of categories over budget.

Required calculation:

Variance = Actual Expense − Budget

How it solves the problem: Once planned limits are entered, the comparison highlights categories requiring attention. It allows users to adjust budgets or reduce spending where necessary.

Problem 8: How many unusual expenses were identified, and what is their total amount?

Problem statement: Unusually large transactions can be difficult to identify when reviewing thousands of records manually.

Solution: Python preprocessing flags unusual expenses using the interquartile range (IQR) method. Power BI then summarises the flagged transactions.

Charts/visuals used:

- KPI card — Unusual Transaction Count.
- KPI card — Total Unusual Expense Amount.
- Bar chart — unusual expenses by category.
- Table — date, category, amount, and transaction description for investigation.

Data insights:

Indicator	Recorded result
Unusual transactions	265
Total unusual expense amount	Approximately ■7,74,835.67
First quartile (Q1)	■30.00
Third quartile (Q3)	■251.00
Upper threshold	■582.50

How it solves the problem: KPI cards show the scale of the flagged transactions, while the table helps users inspect individual records.

Problem 9: Which categories contain the most unusual-expense transactions?

Problem statement: Even after identifying unusual transactions, users need to know which categories contain the greatest number of flagged records.

Solution: The Expense Intelligence page groups unusual transactions by category and ranks the categories by flagged transaction count.

Chart used: Horizontal bar chart

- Y-axis: `Category`
- X-axis: Count of unusual transactions
- Filter: `Unusual_Expense` = True/Yes

Data insights:

Category	Unusual transaction count
Household	49
Transportation	40
Family	30
Subscription	27
Other	26
Food	21
Health	18
Gift	15
Maid	12
Apparel	10

How it solves the problem: The ranked chart helps users prioritise which categories to review first. Pairing it with a transaction-level table lets users check whether the flagged records reflect one-off purchases, recurring expenses, or transactions that need correction.

Problem 10: How much could be saved by reducing discretionary spending?

Problem statement: Users may know that discretionary spending can be reduced but need a practical way to estimate the possible monthly savings from a chosen reduction.

Solution: The Expense Intelligence page includes a scenario-analysis feature that applies a selected percentage reduction to a spending baseline.

Charts/visuals used:

- What-if parameter or percentage selector.
- KPI card — Estimated Monthly Savings.
- Bar chart — current spending versus scenario spending.

Scenario examples recorded in the project plan:

Scenario	Reduction	Illustrative monthly saving
Discretionary shopping	20%	Approximately ■1,600
Food delivery	15%	Approximately ■900

Calculation:

Estimated Savings = Baseline Discretionary Spending × Reduction Percentage

For example, a 20% reduction applied to a ■8,000 monthly shopping baseline gives:

■8,000 × 20% = ■1,600

How it solves the problem: Instead of simply displaying historical expenses, the dashboard allows users to explore a potential change and see the corresponding estimated savings.